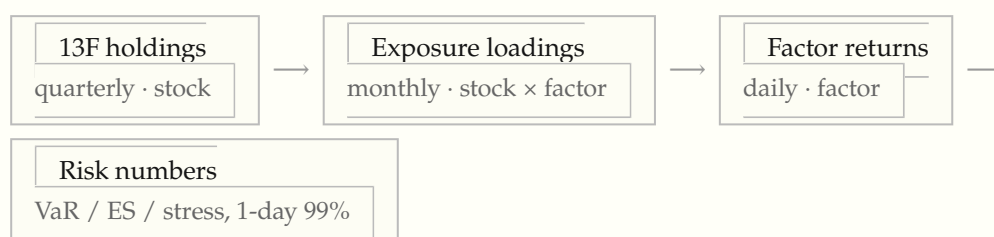


Factor Risk Model — Data & Transformation

Reference

SUMMARY

- **What.** A two-block linear factor model — 10 style factors + a market intercept (the systematic block) and a diagonal name-specific block — over Soros Fund Management's 13F US-equity book (792 names across the sample, 105 held at 2024-12-31).
- **Data, all free / public.** SEC EDGAR 13F (positions), SEC XBRL company facts (fundamentals, point-in-time), OpenFIGI (CUSIP→FIGI identity), Stooq / Yahoo (daily prices). One builder writes six parquet frames; the Atoti cube reads those and nothing else. No factor-return series is bought or downloaded — they are estimated in-house (§3).
- **Loadings.** Each name's style loading is a cross-sectional z-score of a characteristic — Size / Value / Earnings Yield / Leverage / Growth from fundamentals, Beta / Residual Vol / Momentum / Liquidity from prices — standardised by median / MAD on the estimation universe (winsorised ± 3 there, left uncapped on coverage so off-index names read their true loadings).
- **Risk.** Every scenario is the same calculation $dPnL = \sum_k x_k \cdot \Delta f_k$ — only the source of the shock vector changes (historical simulation, event replay, or a hypothetical stress). All figures are 1-day, 99%; book specific vol is currently 0.38%.
- **Descriptors, not factors.** Sector (SIC→GICS) and Country (SEC state of incorporation; ~21% of the book reads non-US) are tags only — there is **no country or industry factor**, so that exposure is carried by the style / market / specific loadings, not a country bet.
- **Cadence.** Three different clocks. **Quarterly** 13F holdings set the book. Each **month-end** the exposure loadings are rebuilt — from as-of fundamentals and prices (§3a–b). The daily stock returns then regress on those month-old loadings to give **daily** factor returns (§3c), which drive every risk number. Quarterly → monthly → daily.
- **Watch-outs.** Monthly loadings drive daily returns (intra-month drift); the estimation cross-section is US-large-cap-tilted; 13F is quarterly, lagged and long-only; specific-risk coverage is partial. Full list in §9.



Fundamentals (per filing, as-of) and daily prices feed the monthly loadings step. The rest of this document is the audit trail behind each line above — every input, every transformation, every check.

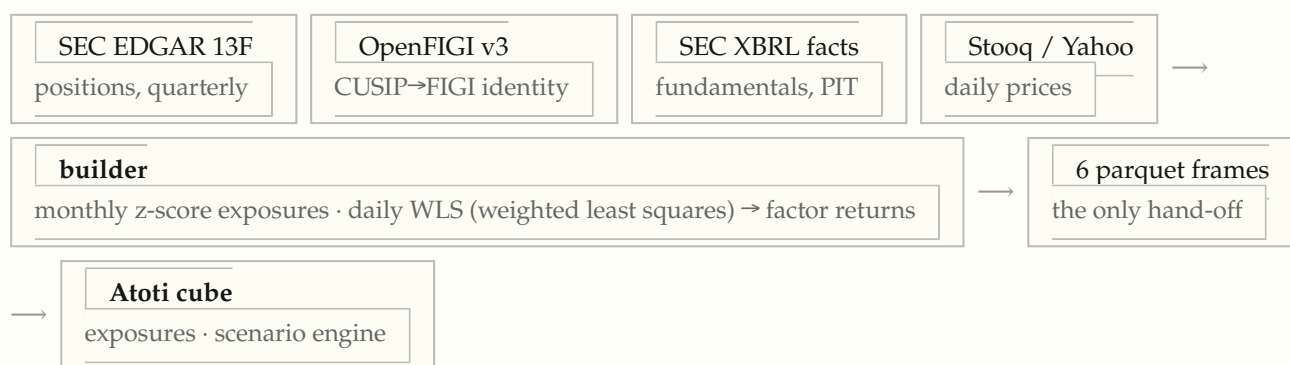
This document is the audit trail for every number the model produces: where each input comes from, what we do to it, and which checks watch it. The book is **Soros Fund Management's US equity positions**, taken from their quarterly SEC **13F filings** (CIK 1029160). The model is a **two-block linear factor model**. Portfolio P&L splits into 10 style factors plus a market intercept (the systematic block) and a diagonal name-specific block. Every scenario is the same calculation,

$$dPnL = \sum_k x_k \cdot \Delta f_k, \text{ where } x_k \text{ is the book's net exposure to factor } k \text{ and } \Delta f_k \text{ is a shock vector.}$$

The only thing that changes between scenario types is where that shock vector comes from: historical simulation, event replay, or a hypothetical stress. **All risk figures are 1-day, 99%.**

PIPELINE AT A GLANCE

The whole pipeline, end to end. Free public feeds go into one builder. The builder writes six parquet frames. The cube reads those frames and nothing else. The builder and the cube share no state in memory, so the six frames are the entire hand-off between data and risk.



Exposures are built **monthly**. The daily factor returns that drive every risk number are **estimated from them in-house** (§3). No factor-return series is bought or downloaded. The monthly-to-daily step is the heart of the model and is set out in §3c.

1 • DATA SOURCES

Where does the factor model come from? Read this first. The factor model is not downloaded from anyone. There is no MSCI or Barra licence and no external factor-return file behind these numbers. The 10 style factors and the market factor are *built here, from scratch*, in three parts:

1. **Factor definitions (the taxonomy), open-source and academic.** What Size, Value, Momentum, EarnYield, Leverage, ResidVol and the rest mean follows the public literature: the Fama–French / **Ken French Data Library** factors, the **BARRA USE4** style set, and the **JKP** characteristic library (Jensen–Kelly–Pedersen, jkpfactors.com). These are conventions, not a data feed. We re-implement the formulas (§3a). We do not import them.
2. **Exposures (loadings), computed here** from SEC fundamentals and Stooq/Yahoo prices as cross-sectional z-scores, monthly (§3a–b).

3. **Factor returns, estimated here, daily**, by regressing each day's stock returns on the latest monthly exposures (§3c–e). This is the standard BARRA cross-sectional approach. The factor returns are an output of our regression, not a bought or downloaded series. The monthly-to-daily step is in §3c.

The only raw data we use (all free) are 13F holdings, identity mappings, company fundamentals and daily prices. The factor structure is built from those, which is why §3 is the substance of the model.

Every input is free or public-domain, fetched over HTTP and cached to disk, so reruns are cheap and do not hit rate-limited APIs again. The four raw feeds below go into the builder. The fifth row, factor returns, is the estimated output from the callout above, listed here so all six frames sit in one place.

INPUT	SOURCE & KEY	CADENCE	GRAIN	WHAT WE RELY ON IT FOR	KNOWN CAVEATS
Positions	SEC EDGAR 13F-HR information tables, CUSIP-keyed (CIK 1029160)	Quarterly, ~45d lag	Stock	The book, held as a weight overlay (§2)	Long US equity only; options (putCall) and bond lots (PRN) are dropped; shorts and intra-quarter trades invisible; public by construction
Identity	OpenFIGI v3 mapping API (CUSIP→FIGI→ticker) + SEC <code>company_tickers.json</code> (ticker→CIK)	On rebuild	Stock	One canonical id (FIGI) joining every frame	Unmapped CUSIPs drop out of the universe; FIGI = Position = SecId everywhere downstream
Fundamentals	SEC XBRL company-facts API, CIK-keyed; tags: Assets, Liabilities, StockholdersEquity, NetIncomeLoss, shares from <code>dei:EntityCommonStockSharesOutstanding</code>	Per filing (point-in-time, as-of joined on <code>filed</code> date)	Stock	Size, Value, EarnYield, Leverage, Growth descriptors	Tag coverage varies by filer; ~no look-ahead by construction; see incident note in §5
Prices	Stooq daily CSV (ticker-keyed) with Yahoo chart-API fallback (split-adjusted close, volume); market proxy = <code>SPY</code>	Daily	Stock	Stock returns (the regression's left-hand side), Beta, ResidVol, Momentum,	Free, not redistributable; Stooq serves a JS anti-bot page to some hosts (fallback handles this transparently)

				Liquidity, market cap	
Exposure loadings	Not sourced, computed. Cross-sectional z-scores of the fundamental + price descriptors, standardised on the estimation universe (§3a–b)	Monthly (rebuilt each month-end)	Stock × Factor	The book's net factor exposure, and the right-hand side of the daily factor-return regression	Month-old when the daily returns regress on them — intra-month drift (§9)
Factor returns	Not sourced, estimated. Daily cross-sectional WLS regression of stock returns on the latest monthly exposures (§3). 10 style factors + 1 market intercept.	Daily, 2,203 trading days (2016–2024)	Factor	The entire systematic block and scenario engine	Quality scales with universe breadth; no vendor benchmark to validate against

A note on cadence and grain. The feeds arrive at different rhythms — positions quarterly, fundamentals per filing, prices daily — but the model runs on a **monthly** calendar: **exposure loadings are rebuilt each month-end** (§3a–b), and the factor returns are estimated **daily** off those month-old loadings (§3c). The *Grain* column is the other axis — every input is per-**stock**, the loadings are per-**stock × factor**, and the daily regression collapses them to per-**factor**. That collapse is §3, the substance of the model.

2 • TRANSFORMATION REGISTER — POSITIONS & IDENTITY

- 13F parsing.** Every information table for the CIK is parsed to (report date, filing date, CUSIP, \$ value). Multiple lots or share classes of one issuer are *summed* per filing.
- Identity resolution.** CUSIP to FIGI via OpenFIGI, ticker to CIK via SEC. All frames key on FIGI. CUSIP, ticker and CIK never leave the builder.
- Point-in-time weight overlay.** Weights are normalised *within each filing* ($\sum w = 1$). Each calendar month-end is then matched to the *latest filing on or before it*, so the book only counts once it has been filed. Names dropped from the newest filing expire with it, so no stale positions persist. Weights sum to exactly 1.0 on all 108 month-ends.

2•B • ESTIMATION UNIVERSE & ITS INTERSECTION WITH THE 13F FILINGS

The factor model is estimated cross-sectionally (§3), so each day it needs a *population of names* to regress over. That is the **estimation universe**. It is built in two parts: **the whole 13F book plus a market-index seed**. Across the sample Soros filed **52** quarterly 13F-HR reports holding a median of **166** cash-equity names each (**2,461** distinct CUSIPs in total, options and bond lots dropped). Those resolve to **1067** names, and the universe keeps them *all*, so the book is always a subset of the universe. To make the cross-section a genuine market rather than one manager's holdings, the universe is then **seeded with the S&P 500**: its constituents are unioned in, adding **129** names not

already in the book, for a total estimation universe of **1196**. The result is a market cross-section that fully contains the book, which is what a Barra-style model needs.

POPULATION	NAMES	DEFINITION / ROLE
13F source population	2,461	Distinct cash-equity CUSIPs across all 52 quarterly filings (before identity resolution)
13F-held names (kept whole)	1067	Those CUSIPs resolved to FIGI, all kept so the full book is covered
+ market-index seed (S&P 500)	129	Index constituents not already in the book, unioned in for market breadth
= estimation universe	1196	The names carried in securities
... with usable exposures	1188	Of the universe, those with enough price/fundamental data to carry loadings, the regression cross-section
Held at ≥ 1 month-end	792	Universe names that surface as an actual book weight at some sampled date
Active book per month-end	80 (med)	Names in the latest filing $\cap$ universe on a given date (range 54–119)

How the populations nest. The 1196-name universe is the whole 13F book (1067 names) plus the index seed (+129 names not already held). 1188 of them have usable loadings and enter the daily regression. The held book is always a strict *subset*: 792 names are held at some point in the sample, and on any one date the active book is the names in the latest filing that are also in the universe (median 80, range 54–119). The point worth making: the regression that builds the factor returns (§3) runs over the full 1188-name market cross-section, not just the names Soros holds. So the factors describe a market, and index names Soros has never held still add breadth. The cube prices the full filed 13F book each quarter, and every §6 number reflects that.

Why this shape. Factor-return quality goes up with breadth, and the daily regression skips any date with fewer than 30 valid names, so a wide market cross-section beats the manager's holdings alone. That is why we seed an index. The only real cost is build time: each name needs cold-cache SEC, OpenFIGI and Stooq calls plus a fundamentals pull. Those are rate-limited, cached to disk, and run in parallel across worker threads, so build time grows with the universe. To go wider, point **SEED_INDEX** at a bigger benchmark (S&P 1500 or Russell 3000). The book stays a subset by construction.

2•C • ESTIMATION-UNIVERSE DATA-QUALITY FILTERS

A real estimation universe is defined by data-quality *rules*, not by index membership (Chris, §5.2: "the estimation universe is a statistical sample chosen for clean estimation, not an investable index"). The pre-filter population is the **point-in-time S&P 500** — the only survivorship-free index available on free data (read as-of each month from a constituent change log that keeps

delisted names), and the desk's settled choice over a broader index (**Chris, 2026-06-23**: "go with option (1) ... survivorship bias should be avoided at all costs"). Each member is then run, point-in-time, through a fixed filter stack and tagged with the *first* stage that drops it:

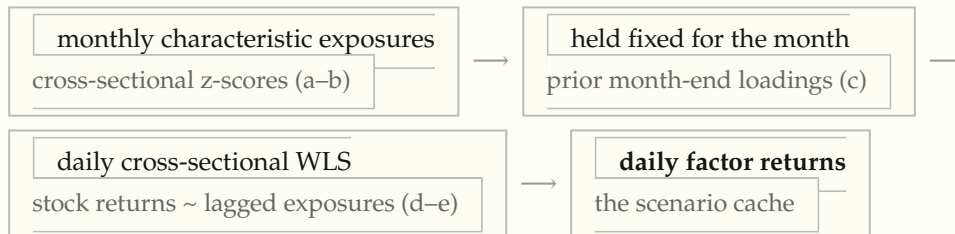
FILTER (IN ORDER)	THRESHOLD	WHAT IT SCREENS / WHY
1 · Listing / security type	primary common	Drop warrants, preferred, units, closed-end funds. S&P 500 membership already implies a primary common listing, so this stage is effectively pass-through here.
2 · Size — min market cap	\$100M	close × shares-outstanding, point-in-time (shares as-of the SEC filed date). Data quality correlates with size; small-cap moves are dominated by microstructure noise.
3 · History — min trading days	252d	Length of the daily price series as-of the month-end — a minimum trading history before admission.
4 · Trading frequency	≥ 90%	Fraction of recent days with non-zero volume — screens stale / thinly-traded quotes.
5 · Liquidity — min ADV	\$1M/day	Trailing mean dollar volume (close × volume). Ensures each day's price is market-clearing, not a stale quote.
6 · Completeness	≥ 6 of 10	Enough non-null style descriptors present for the name to be represented in the model.
7 · Stability buffer	enter 5th / exit 2th pctl adv	Hysteresis on the liquidity rank across months: a name enters above the high band and leaves only below the low band, so membership doesn't churn in and out at a single threshold.
— Free float	<i>unavailable</i>	No free-data source for float-adjusted shares; shown as a disclosed, inert stage (drops nobody).
— Confirmed-M&A removal	<i>unavailable</i>	Needs deal data — confirmed targets track deal probability, not factors. Disclosed, inert.

On the latest month (**2024-12-31**), of **464** evaluable point-in-time S&P 500 names (**39** more are delisted members we can't measure, tagged *data unavailable*, never counted as a filter drop), **446** survive the stack. The funnel is **near-flat by design**: the S&P 500 is already a committee-curated set, so the filters *confirm* clean data rather than carve much away — exactly as expected (**Chris**: for the most-traded SPX names the daily prices are reliable / market-clearing). A name we cannot measure (a delisted member absent from the built universe, or one missing a share count) is shown as *data unavailable*, never as a filter drop, so the per-stage counts only ever reflect genuine threshold failures. Two of the criteria — **free float** and **confirmed-M&A removal** — have no free data source and appear as inert, disclosed stages so the methodology is documented in full.

Thresholds live in `universe_filters.json` (documented and tunable). This filtration is the Phase-2 diagnostic over the estimation universe — surfaced live in the dashboard's " Estimation universe" panel (per-month population→survivor waterfall, per-stage drop list, and the span / high-confidence check) and detailed in `docs/universe-diagnostics-plan.md`. It is a diagnostic on data quality; it does not alter the §3 cross-section the cube currently fits on.

3 • TRANSFORMATION REGISTER — EXPOSURES, FACTOR RETURNS, SPECIFIC RISK

This is the heart of the model and the part with no external benchmark, so every stage is set out. Exposures, factor returns and specific risk all come out of **one regression**. The monthly exposure panel is the regression's right-hand side (the loadings), and the same regression produces the factor-return cache and the residual variances. They are never spliced from different sources. The flow runs in eight stages and moves the data from a *monthly* exposure cadence to a *daily* factor-return cadence. Stage (c) is the bridge.



How to read this. For each month-end we build a table: one row per stock, one column per metric (Size, Value, Momentum, Leverage and so on). Each cell is that stock's value for the month. Stack the monthly tables and you have a block with three axes: stock, metric, month. Read one stock down the months and that is its monthly history for a metric. Read one month across all stocks and that is the cross-section. Stage (a) fills the table from prices and fundamentals. Stage (b) scores each column, one month at a time. The point to be clear on: the exposure is not the raw value. It is a score for where the stock sits on that metric against the rest of the universe that month, in standard deviations. A Size score of +2 means the stock is two standard deviations bigger than the median name that month. That is what lets us compare Size against Value, and one month against the next, even though the raw numbers (log dollars for Size, a ratio for Value) are on different scales.

A worked example. Take one stock, one month, one metric. Say its raw Value (book equity / market cap) is 0.045 in June. To score it we look at every stock's Value that June, take the middle and the spread, and turn 0.045 into something like +0.7. That just says it is a bit cheaper than the typical name that month. We store +0.7 as the stock's Value loading for June. July runs the same way over July's stocks and gives the next point in the series. So the exposures we feed the model are just the scored version of the per-stock monthly metrics you asked about.

1. **(a) Descriptor construction**, per name, on the monthly month-end calendar. *Price-based* (trailing daily prices): **Beta** = $\text{cov}(\text{stock}, \text{SPY}) / \text{var}(\text{SPY})$ over a 252-day window (needs ≥ 120 observations, else withheld); **ResidVol** = annualised σ of that regression's residual ($\times \sqrt{252}$); **Momentum** = 12-1 total return, $\text{price}_{t-21} / \text{price}_{t-252} - 1$ (the most recent month is skipped); **Liquidity** = log of mean daily dollar-volume over the trailing 63 days. *Fundamental-based* (SEC XBRL, as-of joined on the **filed** date so nothing is known before it was reported): **Size** = $\log(\text{mcap}+1)$; **NonLinSize** = $\log(\text{mcap}+1)^3$; **Value** = book equity / mcap; **EarnYield** = net income / mcap; **Leverage** = assets / equity; **Growth** = period-over-period asset growth. Market cap = split-adjusted close $\times$ shares outstanding (DEI cover-page count). **Market-cap floor**: if a name's mcap falls below \$10,000,000 that month we withhold its fundamental descriptors (set them missing, not zero). A corrupt share count would otherwise turn every ratio into raw dollars (\$5).

Two cadences in one monthly panel. The price descriptors are recomputed each month over a *trailing daily window*, so they move every month as new prices arrive. The fundamental descriptors only step when a *new SEC filing lands* (quarterly, as-of joined on the filing date) and stay flat between filings. So in a typical month a name's Beta, Momentum, ResidVol and Liquidity drift while its Size, Value, EarnYield, Leverage and Growth stay put. The panel refreshes price signals every month and fundamental signals every quarter.

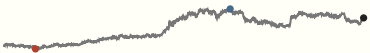
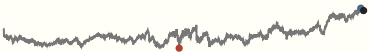
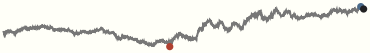
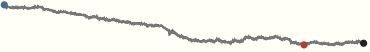
2. **(b) Robust cross-sectional standardisation — the estimation/coverage split.** Each descriptor is standardised *within each month's cross-section*, not across time, robustly: centre by the *median*, scale by the *MAD* ($\times 1.4826$). The centring statistics and the cap come from the **estimation universe only** (the clean S&P 500 seed), not the whole cross-section. **Estimation** rows (the names the factor returns are fit on) are winsorised at $\pm 3\sigma$; **coverage** rows (held names not in the seed) are left **uncapped**, bounded only by a loose ± 10 backstop against corrupt data. Both sets are then re-standardised on the estimation post-clip mean/ σ so they share one scale. This is Chris's data-quality point: a tiny held name far below anything in the S&P 500 reads its *true* large-negative Size loading (≈ -6), not a clip to -3 — the correct statement of a real tilt, and capping belongs only in the estimation set, to protect the regression. If a cross-section has no spread ($MAD \approx 0$) the column is set missing for the month. Median/MAD centring is what makes the cap work: one wrong-units outlier cannot move the centre or scale (§5). `NonLinSize` is also **orthogonalised to Size** (on the estimation fit), so the factor is the genuinely non-linear part of size.
3. **(c) The monthly-to-daily bridge (Barra timing).** Exposures are slow-moving, so we compute them *monthly*. Factor returns have to be daily for the tails to be real. Here is the bridge. Walk consecutive month-ends (d_0, d_1). For every trading day d in the window (d_0, d_1], hold the loadings **fixed at their d_0 (prior month-end) values** and let only the left-hand side, that day's stock returns, change. So one monthly exposure panel feeds about 21 daily regressions before the next panel takes over. This is standard Barra timing: returns lead, loadings lag. It turns ≈ 108 monthly snapshots into **2,203 daily factor-return observations**, which is the difference between a 99% tail estimated from ~ 108 points and one estimated from thousands. The cost: loadings drift inside the month, and the last few days of a window regress on a month-old panel (§7).
4. **(d) Per-day universe and factor screening (removing degenerate factors).** Before each day's regression we screen the design matrix:
 - *Name screen.* Keep names carrying at least **6 of the 10** loadings (the full price block plus some fundamentals), and zero-fill the rest. A missing standardised exposure is the market-average tilt, which is the Barra convention. Requiring complete rows here used to drop whole thin months below the name minimum.
 - *Factor screen.* Drop any factor column whose cross-sectional standard deviation that day is ≤ 0.05 . A near-constant regressor is collinear with the intercept and makes the normal equations near-singular (the old Value/EarnYield blow-ups, §5). That factor simply **does not exist for that day** and contributes no return. The robust z-scoring in (b) should prevent this. The screen keeps the solver safe either way.

- *Minimums*. Skip the day unless at least **30** names survive and the **Size** column is present (Size is also the WLS weight). Daily stock returns beyond $\pm 50\%$ are masked as data errors (split or adjclose glitches) rather than clipped, so a fake -90% never enters a regression.

- (e) The daily cross-sectional WLS regression.** For each surviving day we regress the day's stock returns y on the screened exposure matrix X with a column of ones prepended. That **intercept is the Market factor**. It is weighted least squares with weight $\propto \sqrt{\text{mcap}}$, so large names anchor the fit (implemented by multiplying both sides by $\text{mcap}^{1/4}$). The fitted coefficients *are* that day's factor returns Δf_k , and the residuals are the name-specific returns. The regression is fit on the **estimation universe only** (the S&P 500 seed), so held-but-not-seed names never pull the factor returns. Output: 2,203 days $\times$ (10 style + Market) factor returns, the shared scenario cache.
- (f) Specific (idiosyncratic) risk.** We square each day's regression residual and run it through an EWMA with a **63-trading-day half-life** per name, giving a daily specific *variance* per (name, day). For the cube join we **snapshot this at each calendar month-end** (the model's join cadence). The specific block is strictly *diagonal*: one variance per name, no cross-name specific covariance, in line with the daily VaR horizon. Specific risk is formed for **every coverage name** (its own residual against the estimation-fit factor returns), so the cube prices the whole book even though only estimation names enter the fit.
- (g) Market intercept as a leaf loading.** After the regression we give every (date, name) a **Market** loading of exactly **1.0** in the exposure panel. This is the leaf form of the regression intercept. A fully-invested book (Σ weights = 1) then carries unit market exposure, so the directional market move flows through the scenario engine. We add it *after* stage (e) so the style factor returns are estimated on style exposures only and are not contaminated by the constant.
- (h) Scenario construction.** One table keyed (ScenarioSet, Factor) $\rightarrow$ shock vector. *HistFull* is the full daily factor-return history. *Evt:** is that history cut to a past event window. *Hypo:** is single-day shocks of $n \times \sigma$ per factor (σ = that factor's daily vol). All three feed the same calculation, $dPnL = \sum_k x_k \cdot \Delta f_k$. Only the source of the Δf vector changes.

4 • THE FACTOR BLOCK AS ESTIMATED

FACTOR	GROUP	DAILY Σ	WORST DAY	ON	CUMULATIVE RETURN 2016–2024
Market	Market	1.18%	-12.69%	2020-03-16	
Beta	Style	0.47%	-2.30%	2020-03-09	
EarnYield	Style	0.13%	-0.81%	2020-03-24	
Growth	Style	0.24%	-3.85%	2016-04-22	
Leverage	Style	0.13%	-0.76%	2023-03-13	

Liquidity	Style	0.30%	-1.38%	2020-04-09	
Momentum	Style	0.34%	-4.62%	2020-11-09	
NonLinSize	Style	0.10%	-0.62%	2020-03-25	
ResidVol	Style	0.26%	-1.40%	2022-05-09	
Size	Style	0.28%	-2.12%	2020-03-09	
Value	Style	0.22%	-1.83%	2020-03-09	

Dots: ● minimum, ● maximum, ● latest. The worst Market-factor day is 2020-03-16. The model recovers the COVID crash from estimation alone, with no factor data bought or downloaded.

5 • DATA QUALITY, CONTROLS & THE SHARES-OUTSTANDING INCIDENT

A 30-check DQ suite runs after every rebuild: key uniqueness, referential integrity across all six frames, weight normalisation, calendar continuity, value ranges, and cross-frame coverage.

Current status: **27 pass**, **3 warn**, 0 fail.

Incident, root-caused and defended. During validation the Value/EarnYield factor returns hit $\pm 10^6$ on 21 of 73 regression dates. The cause: `us-gaap:CommonStockSharesOutstanding` facts of *0 shares* (CVNA) and *100 shares* (CHWY) collapsed market cap, which turned ratio descriptors into raw dollars (10^9). Quantile winsorisation could not remove them. The outliers crushed every other name's z-score into a near-constant column, and the regression design matrix went near-singular (condition number 10^8). Four separate defences now stand: (1) shares come from the DEI cover-page tag; (2) a \$10,000,000 market-cap floor withholds descriptors rather than emit wrong units; (3) median/MAD standardisation caps any surviving outlier at a harmless $\pm 3\sigma$; (4) the solver drops degenerate factor columns per date. Each layer is documented at the relevant line of code.

OPEN ITEM (WARN)	DETAIL / ASSESSMENT
held positions have specific_var on the same date	21.1% of (date, position) uncovered
exposures: ≥ 9 of 10 style loadings per (date, position)	min 2, 19.9% below 9
securities: Sector populated	2% 'Unknown' (known stub)

6 • RISK MEASURES & CURRENT BOOK SNAPSHOT (2024-12-31)

Definitions. All on the daily P&L vector of the *current* book (792 names over the sample, 105 held at the snapshot date):

- $VaR\ 99 = -P1(dPnL)$. The 1st percentile of daily scenario P&L (1-day horizon).
- $Worst\ loss = -min(dPnL)$. The single worst scenario day.

- $\text{Specific vol} = \sqrt{(\Sigma w^2 \cdot \sigma^2_{\text{specific}})}$. The diagonal block, currently 0.38% daily, with as-of specific variance available for 79% of held names.
- $\text{Total VaR } 99 = \sqrt{(\text{VaR}^2 + (2.326 \cdot \text{specific vol})^2)}$. The factor tail and an independent-normal idiosyncratic tail, combined.
- $\text{VaR ladder } 95 / 97.5 / 99 = -P5 / -P2.5 / -P1(dPnL)$. The 95/99 spread reads tail fatness; 97.5% is the FRTB regulatory point.
- $\text{ES } 97.5 / 99 = -\text{mean of the worst } 2.5\% / 1\% \text{ of days}$. Expected Shortfall (CVaR) — the average loss in the tail beyond VaR. Coherent / sub-additive where VaR is not, and the Basel FRTB replacement for VaR. *Total ES 97.5* combines it with the idiosyncratic tail in quadrature, like Total VaR.
- $\text{Risk HHI} = \Sigma_{\text{name}} \text{share}^2$, share = a name's fraction of book Total VaR. $1 / \text{HHI} \approx$ the effective number of independent risk bets — the single-number concentration gauge a desk watches against a limit.
- *Marginal VaR / ES* — a member's own P&L on the book's tail day(s). *Additive*: sums to the book VaR/ES, so it splits the tail across factors, sectors, names. *Incremental VaR* — the book VaR released by removing a member (recompute-without); diversification-aware and *not* additive, answering "what does cutting this release?".

NET FACTOR EXPOSURE x_K	LOADING	
ResidVol	+1.133	
NonLinSize	+1.075	
Value	+0.641	
EarnYield	-0.377	
Beta	+0.289	
Liquidity	-0.271	
Momentum	+0.170	
Leverage	-0.059	
Growth	-0.037	
Size	-0.002	

SCENARIO SET	VAR 99	WORST LOSS	TOTAL VAR 99
HistFull (full history)	1.05%	2.82%	1.38%
Evt:COVID2020 (2020-02-01 → 2020-05-31, 82d)	1.93%	2.82%	2.12%
Evt:Rates2022 (2022-01-01 → 2022-10-31, 209d)	1.44%	2.01%	1.70%
Evt:Selloff2018 (2018-10-01 → 2018-12-31, 63d)	0.79%	0.83%	1.19%
Hypo:ValueRotation (Value +2σ, Momentum -2σ)	-0.17%	-0.17%	0.91%

Hypo:RiskOff (Beta -2σ , ResidVol $+2\sigma$, Momentum $+1\sigma$)	-0.38%	-0.38%	0.97%
Hypo:MomentumCrash (Momentum -3σ)	0.17%	0.17%	0.91%

Negative VaR on short event windows means the current factor tilts would have profited in that window's 1st-percentile day.

7 • VAR BACKTEST & MODEL VALIDATION

The 13F book has no live daily P&L, so VaR is validated by a *constant-portfolio backtest*: the current book's exposures applied to the daily factor-return history, rolling a 250-day window to estimate VaR each day and counting *exceptions* where the realised day beat VaR. This validates the methodology, not a trading record.

- *Kupiec POF* — a likelihood-ratio test that the observed exception rate matches the 1% claimed at 99%. $\chi^2(1)$ at 95% = 3.841; above that, the calibration is rejected.
- *Basel traffic-light* — green / amber / red from the binomial CDF of the exception count, generalising the 250-day/99% zones to any window.

Three estimators are available; the default was chosen by a sweep over this book:

ESTIMATOR	TAIL	REACTIVITY	99% BACKTEST
Equal-weight historical sim	empirical (fat)	slow (window edge)	~1.8% — under-covers, amber
EWMA (RiskMetrics, normal)	normal (thin)	fast	~2.1–2.4% — over-breaches, red
FHS (default, $\lambda=0.94$)	empirical (fat)	fast	~1.0% — well-calibrated, green

Filtered Historical Simulation rescales the empirical (fat-tailed) shock distribution by reactive EWMA volatility — reactivity without the normality penalty. Live in the dashboard's VaR-backtest badge.

8 • RISK TOOLING ON THE CUBE

Operational layers exposed in the dashboard, all reading the same cube and frames:

- **Desk limits (RAG)** — book VaR / ES / Risk HHI and single-name / sector weight vs a desk limit set, red/amber/green with breach flags.
- **Data-quality panel** — the §5 checks surfaced live against the served frames, plus the "Unknown" sector fallback for the few names without an SEC SIC.
- **Risk-analyst commentary** — an on-demand written read of any view's numbers (LLM, grounded strictly on the figures shown, no tool or data access), leading with any limit breach.
- **Scoped Q&A (Ask the risk model)** — a free-text question answered by the LLM with *one* tool: `query_cube`, the same guarded pivot the grid uses. It pulls its own slices (each shown

inline) and cites them; it can reach nothing off the allowlist, the filesystem, the web, or any second tool. The agentic loop is bounded.

- **Risk trends** — Scenario VaR / ES and Risk HHI, plus factor exposures, across the 2016–2024 calendar.
- **Stress test** — custom one-day shocks (any per-factor $\sigma \rightarrow$ book P&L, same $\sum x_k \cdot \sigma_k \cdot vol_k$ engine as the Hypo sets) and reverse stress (the single-factor move that breaches a target loss, ranked by vulnerability).
- **Pre-trade / what-if** — resize / drop / add a position and see book VaR, ES, Total VaR, Specific vol and Risk HHI before vs after; the risk math is reproduced in numpy so the "before" reconciles with the cube and the delta is the trade's effect.
- **Drawdown** — the constant-portfolio equity curve and max peak-to-trough over the scenario path (the COVID crash reads $\approx -39\%$), the path lens VaR/ES miss.
- **Liquidity (days-to-liquidate)** — per name $MV / (participation \cdot ADV)$ on a trailing-63-day dollar ADV; the share of the book liquidatable within a horizon, the weighted-average days, the least-liquid names, and any name with no ADV (reported separately). ~87% of the Soros book clears within 5 days at 20% participation.
- **What changed (quarter-over-quarter)** — a deterministic diff between two 13F filings: positions entered / exited / resized, the net factor-exposure drift attributed (rotation vs loading drift), and the book risk delta (VaR / ES / HHI / specific vol, what-if math), with an on-demand written read.
- **Estimation universe (4-phase diagnostic)** — for the held book: which index each name sits in, point-in-time (membership); the DQ filtration funnel over the PIT S&P 500 (§2-c); the span / high-confidence check (is each holding inside the estimation universe's factor space, by Mahalanobis distance); and **style-drift attribution** — the book's net factor exposure over time, with each factor's drift split into rotation (new names) vs held-name loading drift, to read the post-2021 tilt as intentional ($\rightarrow$ benchmark) or not ($\rightarrow$ hedge). See `docs/universe-diagnostics-plan.md`.

9 • LIMITATIONS THE NUMBERS INHERIT

- **13F is a quarterly, lagged, long-only disclosure.** Intra-quarter trading, shorts, options and non-US listings are invisible. The book is the latest filed snapshot.
- **Estimation breadth.** Factor-return quality and stability improve with the number of names in the daily regression (§2-b). A broader cross-section gives more reliable factors.
- **Index-seeded but US-large-cap.** The estimation cross-section is the 13F book plus the S&P 500 (§2-b). That is a real market, but it is US large-cap-tilted, so small-caps and non-US names outside the book are absent. Point `SEED_INDEX` at a broader benchmark (S&P 1500 or Russell 3000) for a fuller market.

- **Sector and Country are free SEC-derived descriptors — neither is a factor.** Sector comes from each filer's SEC SIC code crosswalked to the 11 GICS sectors (CIK-keyed); a handful of BDCs and closed-end funds have no SEC SIC and fall back to "Unknown". Country is the state of incorporation from the same SEC submissions JSON (US state codes → US, foreign codes → the country), so ~21% of the book reads non-US (Alibaba/JD, Canadian energy/industrials, Sanofi). Both are tags only: the model has **no country or industry factor**, so that exposure is carried by the style/market/specific loadings, not a country bet. A residual few ADRs that file a US address with a blank domicile (e.g. Grifols) still read US.
- **Exposures update monthly.** Daily factor returns are regressed on month-old loadings. This is standard Barra timing, but the loadings drift inside the month.
- **Specific-risk coverage is partial.** Names outside the regression cross-section (incomplete descriptors) carry no specific variance that month and add zero to the diagonal block.
- **Free-data licensing.** Stooq and Yahoo price data are free to use but not redistributable. This POC is for internal evaluation only.